🚚 Transportation Problem Solution — Minimum Cost Method

Problem Statement

Consider the transportation network given by the table below and determine the optimal transportation cost, starting the resolution using the Minimum Cost Method.

Problem Data

Supplies (Sources):

Source	Supply
S1	100
S2	800
S3	150
S4	400

Demands (Markets):

Market	Demand
M1	700
M2	250
M3	500

Transportation Costs:

	M1	M2	M3
S1	5	2	8
S2	6	3	7
S3	4	3	6
S4	8	6	4

Step 1: Check for Balance

- Total supply = 100 + 800 + 150 + 400 = **1450**
- Total demand = 700 + 250 + 500 = **1450**

Since supply equals demand, the problem is balanced.

Step 2: Minimum Cost Method Iterations

Iteration 1

- Lowest cost: 2 (S1 → M2)
- Supply S1 = 100, Demand M2 = 250
- Allocate $x_{12} = 100$

	M1	M2	M3	Remaining Supply
S1		100		0
S2				800
S3				150
S4				400
Demand Remaining	700	150	500	

Iteration 2

- Next lowest cost: 3 (S2 → M2)
- Supply S2 = 800, Demand M2 = 150
- Allocate $x_{22} = 150$

	M1	M2	M3	Remaining Supply
S1		100		0

S2		150		650
S3				150
S4				400
Demand Remaining	700	0	500	

Iteration 3

- Next lowest cost: 4 (S3 → M1)
- Supply S3 = 150, Demand M1 = 700
- Allocate $x_{31} = 150$

	M1	M2	M3	Remaining Supply
S1		100		0
S2		150		650
S3	150			0
S4				400
Demand Remaining	550	0	500	

Iteration 4

- Next lowest cost: 4 (S4 → M3)
- Supply S4 = 400, Demand M3 = 500
- Allocate $x_{43} = 400$

	M1	M2	M3	Remaining Supply
S1		100		0
S2		150		650
S3	150			0
S4			400	0

Demand Remaining	550	0	100	

Iteration 5

- Next lowest cost: 6 (S2 → M1)
- Supply S2 = 650, Demand M1 = 550
- Allocate $x_{21} = 550$

	M1	M2	M3	Remaining Supply
S1		100		0
S2	550	150		100
S3	150			0
S4			400	0
Demand Remaining	0	0	100	

Iteration 6

- Next lowest cost: 7 (S2 → M3)
- Supply S2 = 100, Demand M3 = 100
- Allocate $x_{23} = 100$

	M1	M2	M3	Remaining Supply
S1		100		0
S2	550	150	100	0
S3	150			0
S4			400	0
Demand Remaining	0	0	0	

Final Allocation Table

Source	M1	M2	M3	Supply
S1	-	100	-	100
S2	550	150	100	800
S3	150	-	-	150
S4	-	-	400	400
Demand	700	250	500	

Total Cost Calculation

Cálculo do Custo Total:
$Z = (100 \times 2) + (150 \times 3) + (550 \times 6) + (100 \times 7) + (150 \times 4) + (400 \times 4)$
$Z = 200 + 450 + 3300 \div 700 \div 600 + 1600$
$Z = 6850$

Custo total mínimo:

$Z = 6850$

Resposta final:

O custo mínimo de transporte é 6850, utilizando o Método do Custo Mínimo.

Note

The Minimum Cost Method provides a feasible initial solution but does not guarantee optimality. To verify optimality, it is recommended to apply methods such as:

- MODI Method (Modified Distribution Method)
- Stepping Stone Method